

Aiden Van Greuning

📞 (647) 391-1230

✉ aiden.vangreuning@gmail.com

🌐 aiden-vangreuning.github.io/aiden-vangreuning

in [aiden-van-greuning-512944290/](https://www.linkedin.com/in/aiden-van-greuning-512944290/)

Skills

Biomedical Knowledge	Biomechanics, biomaterials, cardiovascular mechanics, biomedical signal processing, tissue engineering, radiation therapy physics, rehabilitation engineering, BioMEMS, medical imaging (X-ray, CT, MRI, PET)
Instrumentation & Laboratory	ECG/EMG/EEG acquisition and analysis, signal conditioning, sensor characterization, tensile/hardness/fatigue/impact testing, XRD, DSC, cell culture (CL2), aseptic technique, biosafety protocols, microscopy
Tools & Communication	MATLAB, Arduino IDE, IEEE-format technical writing, literature review, engineering design documentation, oral presentation, FDA/regulatory overview

Education

2020–2025 **Bachelor of Engineering in Biomedical Engineering**, *Toronto Metropolitan University*

Academic Projects

Capstone: Low Cost Cerebrovascular Flow Pump (Sep. 2024–Apr. 2025) Grade: 89%	• Solely executed a capstone project typically assigned to a four-person team, designing and building a pump system (cost ~\$600, ~75% less cost compared to similar systems in professional labs) to replicate brain arterial blood flow for interventional neuroradiology training • Integrated with a 3D-printed Circle of Willis model, developing an Arduino-based control system for precise flow regulation and collecting flow/pressure data with a flow sensor and Poiseuille's Law using a glycerol–water blood analog
Signal Processing of Cerebromicrovascular Disease (Nov.–Dec. 2024) Grade: 75%	• Processed and analyzed electrocardiogram (ECG) signals from elderly diabetic patients with cerebromicrovascular disease, applying the Pan–Tompkins Algorithm for heartbeat event detection and feature extraction/analysis • Identified PQRST points and achieved 66.7% sensitivity and 62.3% validation accuracy in machine learning classification; ~20% above literature benchmarks

Experience

Apr.–Apr. 2025	Deputy Returning Officer , <i>Elections Canada</i> , Richmond Hill, ON • Verified voter ID, issued ballots and maintained accurate records for 103 electors; assisted with ballot counting, documentation and compliance reporting
Aug. 2024–Mar. 2025	Vice President of Administration (Volunteer) , <i>IEEE EMBS TMU Chapter</i> • Led and delegated tasks to a 7-member team, fostering collaboration with TMU student groups and external professionals to advance joint initiatives
Sep.–Dec. 2023	Innovation Boost Zone Apprentice , <i>PINION (Innovation Boost Zone Startup, TMU)</i> • Recruited volunteers through surveys/outreach and developed the startup's first website to support international student onboarding

Certifications

2025	GCP–Basic; R for Data Science (Analysis & Visualization); TCPS 2: CORE (2022) , <i>N2/CITI Program; LinkedIn Learning; Government of Canada</i>
2026	Medical Devices; Health Canada Division 5; Lean Six Sigma Foundations , <i>N2 Canada; N2/CITI Program; LinkedIn Learning</i>